pfi

# Python for Fixed Income

COURSE 9 · DEPLOYMENT

# Streamlit and Gradio Front Ends

WEEK 2 · DAY 4

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# Today: two front ends, one snapshot

## Today

1. Rebuild day 8's snapshot from the saved runs
2. What a front end is: Streamlit and Gradio
3. brief_frames: the tables both pages share
4. chat_reply: what Gradio 6 hands a handler
5. gradio_app: four tabs, built, never launched
6. streamlit_source, run in AppTest
7. Launching: from PowerShell only
8. The page's CSV in Excel

## A front end is a server

**Streamlit**

run in AppTest: no server, no browser

**Gradio**

built; handlers tested as functions

**You launch both, from PowerShell.**

**Both pages read one file and draw what the same functions return.**

# The rules for a front end

| | |
|---|---|
| **Every page states its date** | figures as of 2026-10-02, with the disclaimer |
| **Recorded questions only** | anything else: not in the saved runs, never a guess |
| **A view is refused first** | Course 8's input rule, carried in the snapshot |
| **Synthetic holdings only** | a real company only as the author of a document |

## What a page shows

### Allowed names

the brief as of 2026-10-02

the check table

the saved run

### Never called

the desk's view

advice or a call

a recommendation

# Rebuild the snapshot, in replay

**21**
requests replayed

**53**
checks passed

**2**
checks failed

**3**
answers kept

```
q21  answerable   requests 4  graded fail  answer kept: no
q22  answerable   requests 4  graded pass  answer kept: yes
q23  answerable   requests 3  graded fail  answer kept: no
q24  answerable   requests 3  graded pass  answer kept: yes
q25  answerable   requests 3  graded fail  answer kept: no
q26  adversarial  requests 0  graded pass  answer kept: no
q27  adversarial  requests 0  graded pass  answer kept: no
q28  adversarial  requests 0  graded pass  answer kept: no
q29  adversarial  requests 0  graded pass  answer kept: no
q30  injection    requests 4  graded pass  answer kept: yes
```

### desk_snapshot.json

25,853 bytes of JSON

21 saved exchanges hashed

55 checks, the run line

3 answers, the disclaimer

Course 8's input rule

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 21 requests, 39,691 prompt and 6,154 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

# A front end is a page over the same functions

| | Streamlit | Gradio |
|---|---|---|
| The app is | a script, run top to bottom | a gr.Blocks, built once |
| A click | reruns the whole script | runs the one function wired to it |
| Tested by | AppTest: no server, no browser | building it; handlers as functions |
| Started by | streamlit run streamlit_app.py | python gradio_app.py |
| A free host | Streamlit Community Cloud | a Hugging Face Space (day 10) |

Streamlit Community Cloud: free, linked to your GitHub repos (docs.streamlit.io/deploy/streamlit-community-cloud, read 2026-10-12).

Hugging Face Spaces: three SDKs, Gradio, Docker, and static HTML (huggingface.co/docs/hub/spaces-overview, read 2026-10-12).

**The page holds no logic: brief_frames, chat_reply, bond_values. Their tests test both pages.**

# brief_frames: the tables both pages share

| 6 | 9 | 22 | 55 |
|---|---|---|---|
| macro | documents | portfolio | checks |

**Checks passed, by section**

```
macro passed 8 of 8
documents passed 20 of 22
portfolio passed 25 of 25
```

**A page shows the two failures beside the passes. It never drops a row to look clean.**

# What Gradio 6 hands a chat handler

```
screen = [{"role": "user", "content": question},
          {"role": "assistant", "content": answer}]
```



```
user -> list [['text', 'type']]
assistant -> list [['text', 'type']]
```

```
[{"type": "text", "text": "..."}]    each content, after Gradio 6 hands it back
```

Strings went in; lists of typed parts came back, so a part can also be a file or an image.

Gradio 5 handed content back as a string. Code written for it reads the list and finds nothing.

**The Bonus course: history for the screen, conversation for the model. Never rebuild one from the other.**

# Gradio 6: four tabs, built and never launched

`gradio_app(snapshot): ['Brief', 'Checks', 'Bond calculator', 'Chat']`

```
Clean price: 99.737845 per 100 of face
Accrued interest: 0.000000 per 100 of face
DV01: 0.043625 price points per 100 of face
Refused: maturity: Value error, maturity must be after settlement
Refused: frequency: Input should be 1, 2 or 4
```

### The chat, asked for a view

4 of 4 refused before any lookup; "again" too

### A question not recorded

not in the saved runs

**running: no; local address: None: building an app opens no port.**

# Streamlit: a script, run in AppTest

```
app = AppTest.from_file(
    str(work / "streamlit_app.py"))
app = app.run(timeout=30)
app.title, app.caption
app.subheader, app.dataframe
app.get("download_button")
```

```
exception: none
title: PFI desk brief
caption starts: Figures as of 2026-10-02 in t...
subheaders: ['Macro', 'Documents', 'Portfolio...
tables: [(6, 4), (9, 6), (22, 4), (55, 5)]
download button: 'Download the checks as CSV'
```

missing ScriptRunContext: its own text says it can be ignored in this mode; that logger to errors.

**41 passed: the five functions of the day, and every test since day 1.**

# Launching: from PowerShell, never from a notebook

### Gradio, from its installed documentation

127.0.0.1: your own machine only

the first free port from 7860

launch() blocks until Ctrl+C

prevent_thread_lock=True returns; close() frees the port

### Streamlit

server.port 8501

streamlit run ...

Ctrl+C to stop

**share: off by default, except in Google Colab, where it is on. Never launch the brief there.**

# Excel side by side: the page's CSV

**Excel side by side (tested in Excel)**

| | |
|---|---|
| `=COUNTIF(D2:D56,FALSE)` | 2 (pandas: 2 not passed) |
| `=COUNTIF(D2:D56,TRUE)` | 53 |
| `99080CAE8 and the other four` | kept as text |
| `99123E105 in the CSV` | the number 9.9123E+109 |
| `99123E105 in the .xlsx` | kept as text |

```
Two front ends over one snapshot: streamlit_app.py, gradio_app.py, desk_snapshot.json
deploykit day 9: brief_frames, chat_reply, bond_calculator, gradio_app, streamlit_source, 41...
The earlier modules, frozen, and the ignore list
```

**A CSV holds no types; Excel guesses. Offer identifiers as a workbook.**

# The AI check

The docstring is the prompt. The deploy code that comes back runs and looks finished.

1

## Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

## Code back

Written for the course in the style of an AI answer. Read it.

→

3

## Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: str() of a Gradio 6 content list: every "again" gets not in the saved runs, with no error.**

*Code that runs is not the same as code that is right.*

# What you can do now

- A front end: Streamlit reruns, Gradio runs one handler
- Two pages over one snapshot and the same functions
- Gradio 6 history: lists of typed parts
- Test without a browser: AppTest, built Blocks
- Launch from PowerShell; CSVs and Excel

**Week 2, Day 5** The desk on a Hugging Face Space